NirogScan Comprehensive Health Report

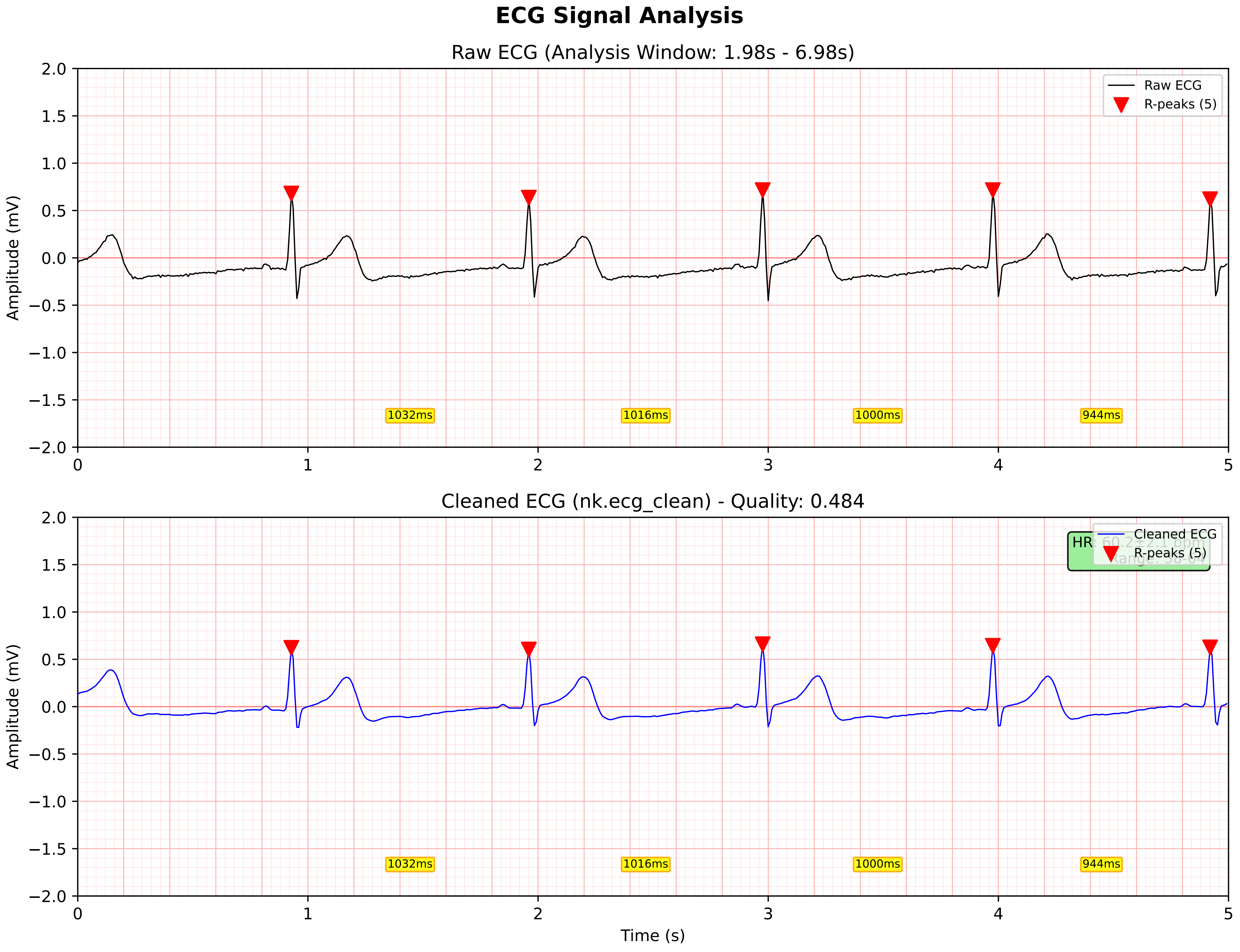
Patient Information

Name:	ak
Age:	24
Gender:	Male
Weight:	N/A
Height:	N/A
Notes:	None

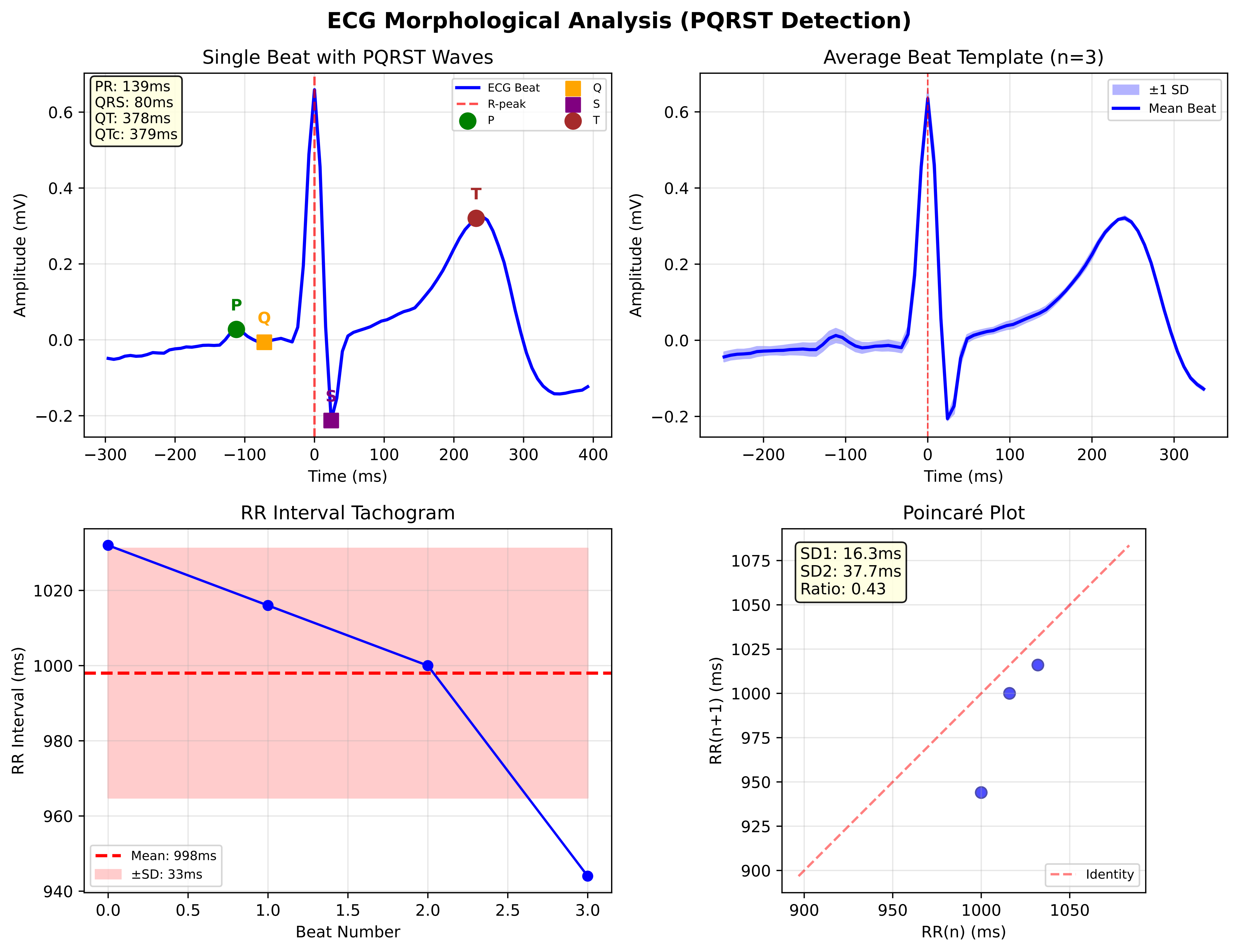
Recording & Analysis Summary

ECG Duration:	11.8s (1475 samples)
ECG Sampling Rate:	125 Hz
Best Analysis Window:	1.98s - 6.98s
Window SNR:	20.6 dB
R-peaks Detected:	5
Analysis Engine:	NeuroKit2 v0.2.12
ECG Heart Rate:	60.2 ± 2.1 bpm
PPG Duration:	23.6s

Report Generated: 2026-01-31 19:27:04



ECG Signal Analysis
Raw ECG (Analysis Window: 1.98s - 6.98s)
Raw ECG
R-peaks (5)
1032ms
1016ms
1000ms
944ms
Amplitude (mV)
Cleaned ECG (nk.ecg_clean) - Quality: 0.484
HR: 60.2±2.1 bpm
Range: 58-64
Cleaned ECG
R-peaks (5)
1032ms
1016ms
1000ms
944ms
Amplitude (mV)
Time (s)



ECG Morphological Analysis (PQRST Detection)
Single Beat with PQRST Waves
PR: 139ms
QRS: 80ms
QT: 378ms
QTc: 379ms
ECG Beat
R-peak
P
Q
S
T
Amplitude (mV)
Time (ms)
Average Beat Template (n=3)
±1 SD
Mean Beat
Amplitude (mV)
Time (ms)
RR Interval Tachogram
Mean: 998ms
±SD: 33ms
RR Interval (ms)
Beat Number
Poincaré Plot
SD1: 16.3ms
SD2: 37.7ms
Ratio: 0.43
Identity
RR(n+1) (ms)
RR(n) (ms)

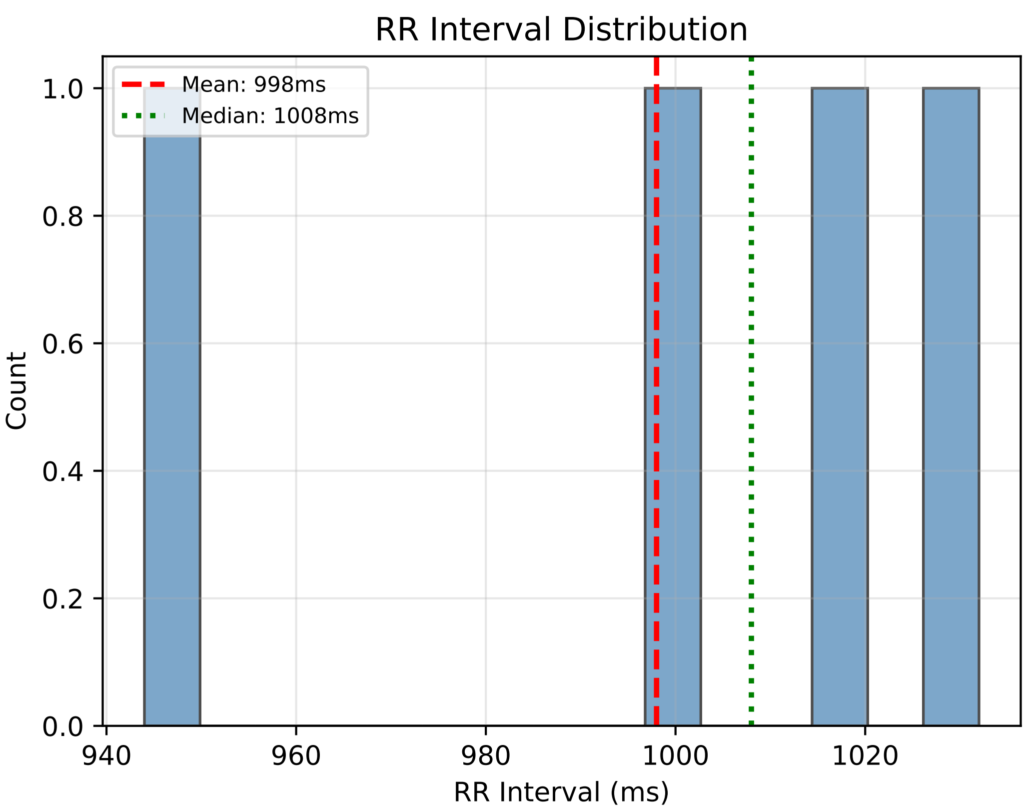
HRV Analysis - Time Domain

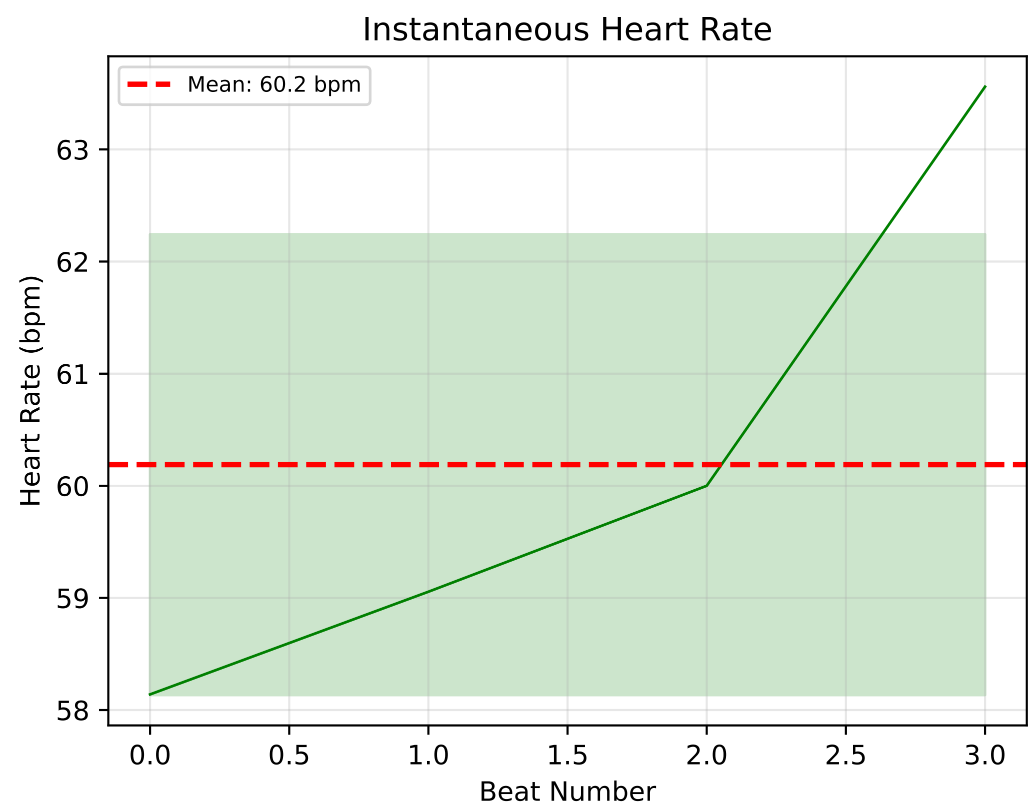
Core Time Domain Metrics

Average RR Interval:	**998.00**
Heart Rate Variability:	**38.30**
Short-term HRV:	**34.87**
Percentage of Large Changes:	**25.00**
Percentage of Small Changes:	**25.00**
Successive Differences SD:	**23.09**

Additional Time Metrics

Median RR Interval:	**1008.00**
Median Absolute Deviation:	**23.72**
Coefficient of Variation:	**0.04**
Interquartile Range:	**34.00**
Triangular Index:	**0.00**
HRV Triangular Index:	**4.00**





HRV Analysis - Frequency & Nonlinear Domains

Frequency Domain Metrics

HRV_TP:	**0.00**

Insufficient frequency data

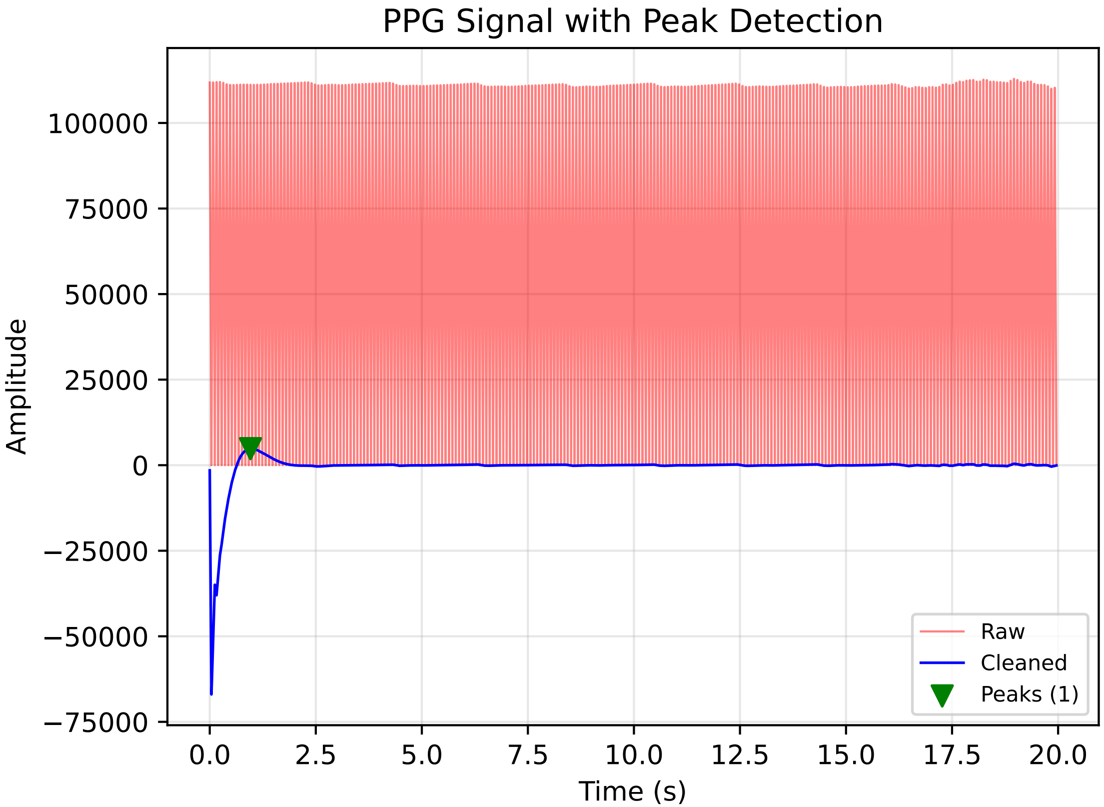
Nonlinear Metrics (Poincaré)

Poincaré SD1:	**16.3299**
Poincaré SD2:	**37.6652**
SD1/SD2 Ratio:	**0.4336**
Poincaré Area:	**1932.2994**
Cardiac Sympathetic Index:	**2.3065**
Cardiac Vagal Index:	**3.9930**

Entropy & Fractal Metrics

Approximate Entropy:	**0.4055**
Sample Entropy:	**-inf**
Fuzzy Entropy:	**3.4816**

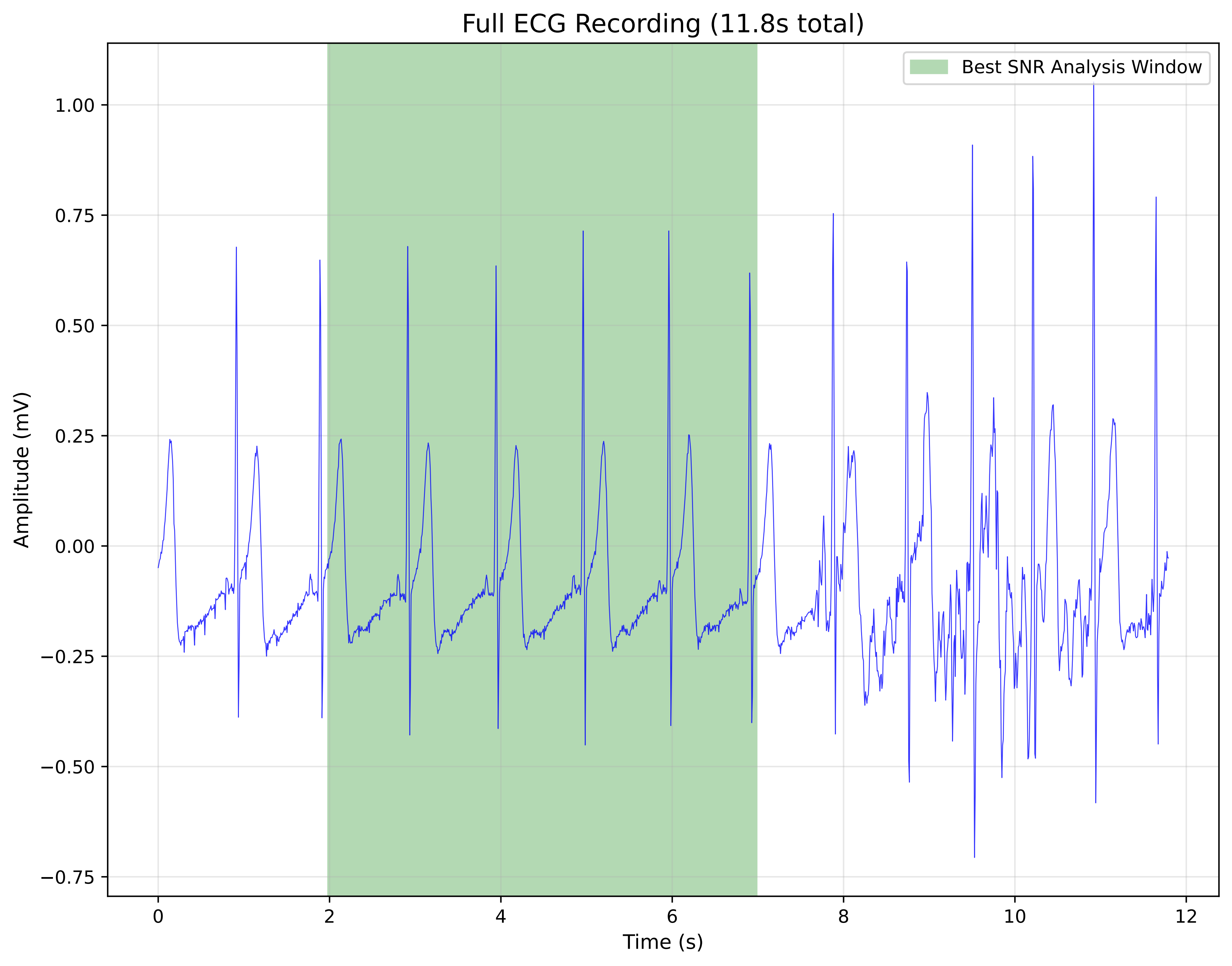
PPG Signal Analysis (MAX30102 IR Channel)



Insufficient HR data

Insufficient data

Insufficient data



Full ECG Recording (11.8s total)
Best SNR Analysis Window
Amplitude (mV)
Time (s)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0
2
4
6
8
10
12

Analysis Summary

ECG Analysis:

Signal Quality (SNR):	20.6 dB
R-peaks Detected:	5
Heart Rate:	60.2 ± 2.1 bpm
SDNN:	38.3 ms
RMSSD:	34.9 ms

PPG Analysis:

Peaks Detected:	1

Device Information:

ECG Sensor:	AD8232 (Gain: 1100)
ECG Sampling:	125 Hz
PPG Sensor:	MAX30102 (Red 660nm, IR 880nm)
PPG Sampling:	25 Hz
ADC:	12-bit, 3300mV ref

Understanding Your Results

(What each measurement means in simple terms)

- **Average RR Interval:**
  The average time between heartbeats in milliseconds. Normal range: 600-1000ms.
- **Heart Rate Variability:**
  How much your heartbeat timing varies. Higher is generally better, indicating a healthy, adaptable heart. Normal: 50-100ms.
- **Short-term HRV:**
  Measures beat-to-beat changes. Reflects your parasympathetic (rest & digest) nervous system activity. Normal: 20-50ms.
- **Percentage of Large Changes:**
  Percentage of heartbeats that differ by more than 50ms from the previous one. Higher values indicate good heart health. Normal: 3-25%.
- **Poincaré SD1:**
  Short-term variability from Poincaré plot. Related to RMSSD.
- **Poincaré SD2:**
  Long-term variability from Poincaré plot. Related to SDNN.
- **Approximate Entropy:**
  Signal complexity/predictability. Lower = more regular heartbeat.
- **Sample Entropy:**
  Similar to ApEn but more consistent. Measures heart rhythm complexity.
- **PR Interval:**
  Time from atrial to ventricular activation. Normal: 120-200ms.
- **QRS Duration:**
  Time for ventricles to contract. Normal: 80-120ms.
- **QT Interval:**
  Total ventricular activity time. Varies with heart rate.
- **Corrected QT:**
  QT interval adjusted for heart rate. Normal: <440ms.

*Note: These measurements provide insights into heart health but are not medical diagnoses.
Please consult a healthcare professional for medical advice.*